frontiers in Computer Science

TYPE Opinion
PUBLISHED 08 September 2026
DOI 10.3389/fcomp.2026.1892907

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OPEN ACCESS

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RECEIVED 27 May 2026
REVISED 18 August 2026
ACCEPTED 24 August 2026
PUBLISHED 08 September 2026

CITATION
Schilhab TSS (2026) Totalization by design: anthropomorphic responsiveness and ethical limits in human–robot interaction.
Front. Comput. Sci. 8:1892907.
doi: 10.3389/fcomp.2026.1892907

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Totalization by design: anthropomorphic responsiveness and ethical limits in human–robot interaction

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KEYWORDS

anthropomorphic design, embodied cognition, ethical responsiveness, functional social responsiveness, human–robot interaction, Levinas, moments of meeting, totalization

1 Introduction

Human-likeness, in the form of anthropomorphic cues, is a central design principle in social robotics, shaping the ways artificial agents are perceived and engaged with (Goldman et al., 2023; Reuter et al., 2025).

Anthropomorphic cues such as facial configurations, vocal modalities, gaze-like orientations, affective expressions, behaviors and verbal interactions mediate the user's perception of attention, intention, emotion, and responsiveness (Hoff and Bashir, 2015; Song, 2020). By aligning machine behaviors with the embodied and relational expectations of human interlocutors, I conjecture that design choices transform users' expectations about agents' understanding, their capacity for caring, and their ability to be relied upon (Carter et al., 2024).

This learning effect is both fascinating and ethically disturbing. In essence, the more convincing the anthropomorphic cues of a system appear, the stronger the cue-driven attribution of agency, and the greater users' inclination to ascribe expectations of reciprocity, empathy, deliberation, and responsibility.

Often, this dynamic is framed as a matter of the human-like agent simply deceiving the user. However, construing the discussion as a matter of false beliefs fails to reveal the deeper issue of what encounters based on anthropomorphic cues establish over time. Social robots are designed to be available on demand and adaptable when encountering friction in interactions, while maintaining predictable performance. When strong cues of agency arise from engineered availability, users may unconsciously learn to expect that such responsiveness, timely, coherent, and reliably tailored to their needs, also pertains to the original encounters between humans they simulate (e.g., Barsalou, 2016).

To articulate the ethical implications of this, I draw on the philosopher Emmanuel Levinas's account of ethics as grounded in encounters in which the other resists being reduced to what can be comprehended, predicted, or possessed (Levinas, 1996/1951, Levinas, 2003). Levinas's name is often associated with "ethics as first philosophy," which prioritizes alterity over the drive toward unity and interiority (Lipari, 2012; Métais and Villalobos, 2022). To Levinas, this encounter is not first cognitive or theoretical but ethical: it calls us into responsibility prior to reflection, categorization, or judgment. In this sense, responsibility for the Other precedes and grounds all other forms of thought. In the encounter, the other

> "presents himself, exceeding the idea of the other in me." (Levinas, 1997, p. 50)

For Levinas (1969), the ethical danger lies in totalization, which involves approaching the Other primarily through concepts, expectations, and categories that render alterity intelligible and manageable rather than allowing it to remain irreducibly other.

Like Dell'Oro (2022), my approach adopts a Levinasian understanding of personhood that rejects the idea that robots can be persons [see also Schilhab (2015a)]. However, their human-like capacity for engaging us might still have consequences for how we commit ourselves ethically in human interactions. Here, the opinion piece proposes a plausible theoretical pathway through which such effects could occur.

Levinas' position is silent about developmental and learning aspects of the encounter. Levinas claims that the ethical relation to the Other is the very condition of possibility for subjectivity itself. Thus, the idea that humans become ethical through development, socialization, or moral education is irrelevant. In Levinas' account, responsibility for the Other precedes freedom, intentionality, and reflective consciousness. This means that the pre-reflective exposure to the ethical demand of the Other is the condition of possibility for, rather than the product of, moral learning (Critchley, 2002; Bernasconi, 1999).

In relation to this interpretation, Katz (2008) argues that moral learning should then be understood as cultivating an awareness of the responsibility that already constitutes the subject. Accordingly, from a Levinasian perspective, what humans might need to learn is to recognize and respond to the ethical responsibility that precedes knowledge, autonomy, and deliberate choice (Katz, 2007, 2008).

The Levinasian question, therefore, is whether increasingly prediction-based expectations risk constraining openness to an Other whose alterity necessarily exceeds anticipation, thereby increasing the risk of totalization (Levinas, 1969, 1996/1951, Levinas, 1998; Lipari, 2012).

This concern resonates with Turkle's (2011, 2015) argument that digital technologies and social robots shape our expectations of human relationships, leading us to "expect more from technology and less from each other." However, whereas Turkle primarily identifies this as a sociocultural phenomenon, the present approach proposes a cognitive mechanism through which such changes may actually occur.

For the purposes of this paper, I introduce the analytical distinction between functional social responsiveness and ethical responsiveness to explain why anthropomorphic cues carry an ethical charge. Whereas functional social responsiveness refers to the desired operation of social cues that sustain further interaction and invite trust, ethical responsiveness refers to responsiveness to an other who cannot be exhaustively anticipated, elucidated, and grasped as a transparent whole. I argue that contemporary human–robot interaction (HRI) predominantly optimizes the former, whereas Levinas' ethics concerns the latter.

To develop this argument, I first draw on the embodied cognition perspective [Maturana and Varela, 1987; Varela et al., 1991; Lakoff and Johnson, 1999; Wilson, 2002; Barsalou, 2008; for an application in HRI research, see Mancusi et al. (2025)] to explain how anthropomorphic cues elicit embodied processes that support agency attribution and socially meaningful engagement. Building on Barsalou's (1999, 2003, 2009) theory of situated conceptualization and Hesslow's (2002, 2012) simulation theory, I suggest that repeated human–robot interactions may become incorporated into the predictive structures through which people perceive, interpret, and respond to subsequent social encounters. I then introduce Stern's framework of moments of meeting (Stern et al., 1998; Stern, 2004) to argue that although interpersonal encounters always unfold against the background of such experience-based expectations, they also retain the potential to transform these expectations through authentic intersubjective engagement. Hence, following Stern, prior experience shapes human interaction without fully determining it: authentic intersubjective engagement retains the potential to disrupt and transform the expectations brought to subsequent encounters. Finally, I consider the implications of these processes for Levinas' ethics.

Figure 1 summarizes the theoretical mechanism proposed in this paper and illustrates how anthropomorphic design may ultimately influence the experiential conditions for ethical encounters.

2 Embodied mechanisms of expectation formation

When an entity appears to attend, respond, and coordinate—that is, appears to be goal-directed and autonomous—we tend to perceive it as an agent rather than a mere object (Dennett, 2008; Schömbs et al., 2023; Shi et al., 2025). The embodied cognition perspective explains this "readability" through the neural mechanisms at play during interaction.

According to Barsalou's (1999, 2003, 2009) theory of situated conceptualization, people's understanding of a situation is shaped by the reactivation of previous experiences in similar contexts rather than by abstract knowledge (amodal symbolic representation) alone. Through repeated experiences, perceptual, motor, affective, and social elements that consistently co-occur become increasingly associated with one another, reflecting the Hebbian principle that

> "neurons that fire together wire together." (Hebb, 1949)

Consequently, encountering a familiar situation automatically reactivates a situated conceptualization that generates predictions about what is happening, how others are likely to behave, how one should respond, and what the situation means (e.g., Sheckley and Bell, 2006; Schilhab, 2013, 2015a,b, 2017a). Hesslow's (2002, 2012) simulation theory provides a neurocognitive account of this process by proposing that the brain continuously simulates perception and action based on prior experience. Behavior and meaning therefore emerge from both the current stimulus and the interaction between the present situation and experience-based simulations that shape perception, expectations, and action (Barrett, 2009).

Accordingly, repeated interactions with social robots may gradually become incorporated into individuals' situated conceptualizations of social encounters. This learning transfer

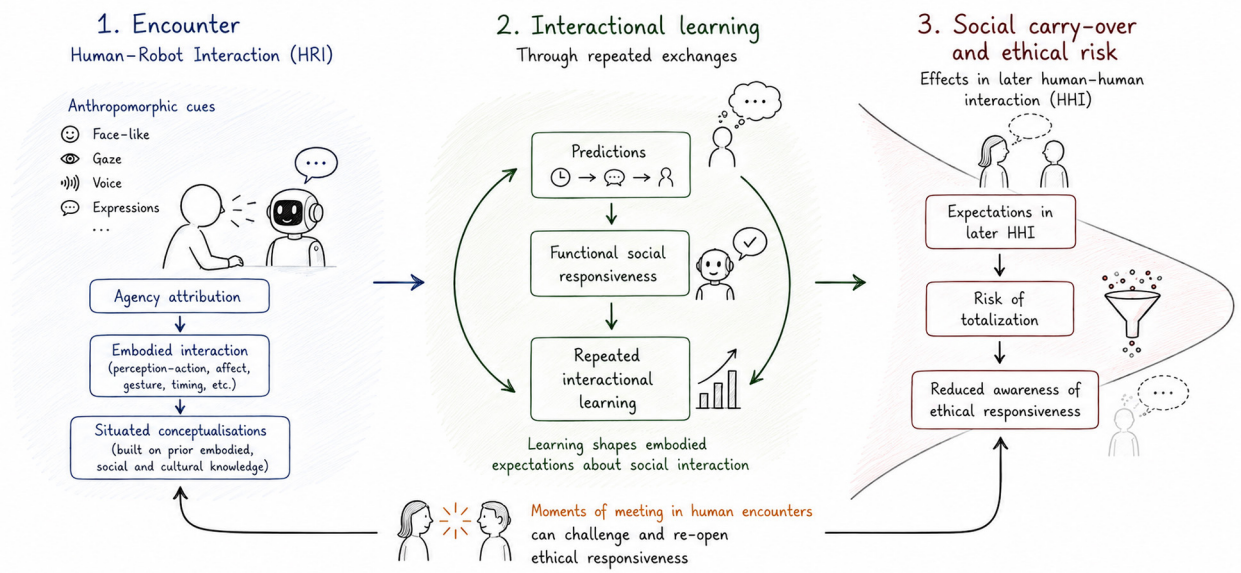


FIGURE 1
Plausible hypothesized pathway through which anthropomorphic cues in human–robot interactions may shape expectations in later human–human interactions and increase the risk of totalization.

arises because situated conceptualizations are organized around perceived social situations rather than around ontological categories, and because the neural processes and perceptions involved are predominantly outside our cognitive control (Barsalou et al., 2003).

As experiences with human-like robots accumulate, they may contribute to the predictions people make in subsequent interactions, including encounters with other humans [for predictions in the neurosciences, see Bar (2009)]. Consequently, experiences acquired through human–robot interaction may influence not only expectations of robots but also the implicit assumptions, behavioral tendencies, and relational expectations that individuals bring to future human–human encounters. This possibility provides a theoretical foundation for investigating whether repeated human–robot interactions can shape the dynamics of later interpersonal encounters.

Daniel Stern's concept of moments of meeting adds to this perspective by describing how interpersonal relationships can be transformed through authentic, mutually responsive encounters (Stern et al., 1998; Stern, 2004). Whereas Barsalou and Hesslow explain how expectations are acquired and generalized across situations, Stern explains how they can occasionally be transformed through authentic interpersonal encounters.

A "*moment of meeting*" emerges when a "*now moment,*" an unexpected and affectively significant disruption of ongoing interaction, is met with a genuine, contingent response from both interaction partners. Such moments transform pre-existing implicit relational expectations. They cannot be reduced to the application of pre-existing behavioral scripts; rather, they create new forms of implicit relational knowing that subsequently influence how individuals engage in future relationships (Stern et al., 1998).

From the perspective of situated conceptualization, however, individuals do not enter moments of meeting encounters without prior expectations. Rather, they bring experience-based predictions about social interaction that shape how they initially perceive, interpret, and respond to another person. Although moments of meeting have the potential to transform these expectations by creating novel relational experiences, the situated conceptualizations developed through previous interactions provide the background against which such moments unfold.

From the perspective of situated conceptualization, individuals inevitably enter moments of meeting with expectations shaped by previous experience. Stern's contribution is precisely that genuinely contingent encounters can reorganize these expectations by creating new implicit relational knowing. This possibility becomes ethically significant from a Levinasian perspective because it suggests that, although openness to the Other depends on expectations, it also depends on those expectations remaining revisable in the encounter itself.

3 Risk of totalization

Now, the pressing issue arises of what kind of "other" is made perceptually available in encounters with human-like robots. Following Levinas (1996/1951, 2003), the ethical encounter is never a meeting with an object that can be fully and conceptually grasped, but a relation in which the other resists reduction to my concepts and my cognitive control. The danger is totalization: the impulse to complete the other by enclosing alterity within what is already intelligible to me (Lipari, 2012). For Levinas, totalization is not merely a theoretical sin. At the level of lived experience, individuals are implicitly drawn toward smoothness rather than interruption, coherence rather than ambiguity, and predictability rather than vulnerability. Such inclinations might be understood in light of

theories of cognitive load and tacit learning (Tricot et al., 2008; Hasse, 2015; Schilhab, 2017a,b).

If such expectations are carried into later moments of meeting, they become ethically significant.

The significance of this possibility is that human–robot interactions increase the likelihood of Instances of totalization that may shape the relational expectations from which moments of meeting emerge and against which transformative interpersonal experiences occur.

This possibility becomes particularly significant when distinguishing between functional social responsiveness and ethical responsiveness. Functional social responsiveness refers to the effective use of social cues that sustain interaction, coordinate behavior, and invite trust. It relies on interactions being sufficiently predictable for participants to anticipate each other's responses. Contemporary HRI design deliberately optimizes for precisely these qualities because they support interactional success, trust, engagement, and user acceptance. Ethical responsiveness, by contrast, concerns responsiveness to an Other whose alterity always exceeds prediction, categorization, and conceptual grasp (Levinas, 1969, 1998). Rather than responding to expected social signals, ethical responsiveness entails remaining open to what cannot be fully anticipated.

Whereas HRI research commonly evaluates successful interaction in terms of trust, rapport, engagement, or interaction fluency, a Levinasian perspective suggests that these qualities primarily reflect functional social responsiveness. Ethical responsiveness concerns a different dimension of social encounter, which is not the successful coordination of interaction, but openness to an Other whose alterity exceeds anticipation, prediction, and conceptualization. Thus, the ethical significance of an encounter cannot be reduced to whether interaction is experienced as smooth, trustworthy, or mutually satisfying, but depends on whether the Other is encountered as irreducibly other rather than merely as a predictable social partner.

In a nutshell, the use of anthropomorphic cues as a design principle has the potential to bolster the former while fostering an illusion of the latter. As a result, agency attribution becomes easier to trigger, while the conditions for genuine alterity become harder to encounter.

4 Concluding remarks

My intention is not to claim that anthropomorphic technologies diminish human morality or make ethical encounters impossible. Functional social responsiveness can reduce anxiety, facilitate communication, and foster trust. The concern is different. Drawing on embodied cognition, this paper proposes a theoretical mechanism through which repeated human–robot interactions may gradually calibrate the expectations people bring to later encounters with other humans.

Social life necessarily depends on prediction and coordination. However, the ethical concern arises if interactions increasingly privilege functional social responsiveness, responsiveness organized around anticipation, fluency, and successful coordination, over openness to an Other whose alterity exceeds prediction.

This possibility also extends Katz's (2007, 2008) interpretation of Levinas. If moral education cultivates awareness of a responsibility that already precedes reflection, then anthropomorphic technologies may influence the experiential conditions under which such awareness is cultivated. The question is therefore how anthropomorphic design can support interaction without narrowing our openness to the irreducible alterity of other human beings.

Author contributions

TS: Writing – original draft, Writing – review & editing.

Funding

The author(s) declared that financial support was not received for this work and/or its publication.

Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that Generative AI was used in the creation of this manuscript. The author used ChatGPT Images 2 (OpenAI; accessed 5 July 2026) to generate and iteratively revise an initial visual draft for Figure 1. The author(s) directed the conceptual design, reviewed and edited the final visualization, verified its consistency with the manuscript, and take full responsibility for its content.

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